TCP Chat Server and Client using Socket Programming in C

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1 Introduction

This project implements a multi-client chat application using TCP socket programming in C.

2 Requirements

- GCC Compiler
- Linux / Ubuntu / WSL
- POSIX Threads Library (`pthread`)

3 Compilation Instructions

```
gcc server.c -o server
gcc client.c -o client
```

4 Execution Instructions

```
./server <server_port_number>
./client <server_ip> <server_port_number>
```

5 Usage

- 0 → Broadcast
- 1 → Private Message
- 2 → Exit

6 Additional Functionalities

- Multi-threaded server
- Concurrent send/receive using threads
- Broadcast messaging
- Private messaging using IP
- Mutex-based synchronization
- Dynamic client join/leave
- Safe termination using shutdown()
- Custom message labeling
- Parsing without strtok

7 Screenshots

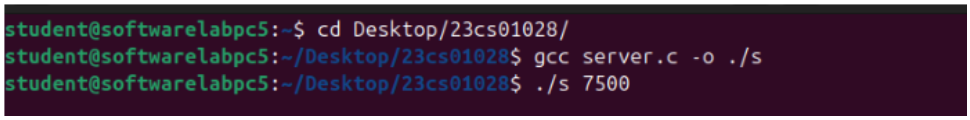

Figure 1: Server Initialization

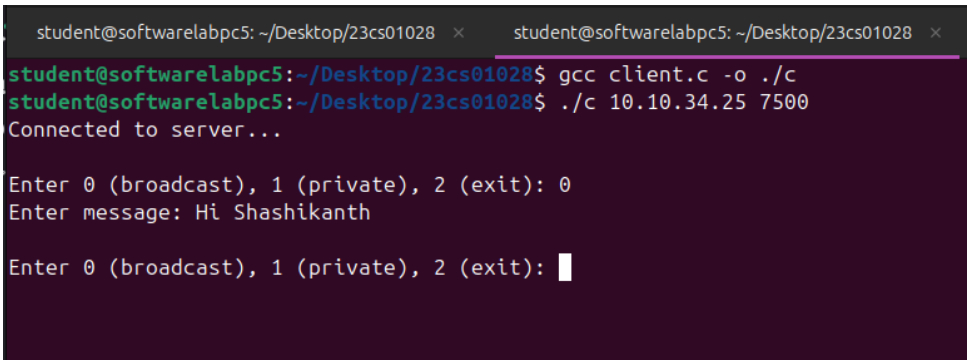

Figure 2: Group Message Sent

Figure 3: Group Message Received

```
Enter 0 (broadcast), 1 (private), 2 (exit): 1
Enter receiver IP: 10.10.34.26
Enter message: Hi, this is Privately sent to You!!

Enter 0 (broadcast), 1 (private), 2 (exit):
```

Figure 4: Private Message Sent

```
[Message]: Message from client 5: Hi, this is Privately sent to You!!
```

Figure 5: Private Message Received

```
Enter 0 (broadcast), 1 (private), 2 (exit): 0
Enter message: hi
```

Figure 6: Group message 'hi' is sent

```
[Message]: [GROUP] Client 6: hi
```

Figure 7: Group message received by one client(out of 2 other than sender)

```
Enter 0 (broadcast), 1 (private), 2 (exit):
[Message]: [GROUP] Client 6: hi
```

Figure 8: Group message received by another client

```
Enter 0 (broadcast), 1 (private), 2 (exit): 1
Enter receiver IP: 10.10.34.26
Enter message: saumay
```

Figure 9: Private message sent

```
[Message]: Message from client 5: saumay
```

Figure 10: Private message received to the client for whom the message was sent

```
student@softwarelabpc5:~/Desktop/23CS01037/CN$ ./c
Usage: ./c hostname port
student@softwarelabpc5:~/Desktop/23CS01037/CN$ ./c 10.10.34.25 7500
Connected to server...
```

Figure 11: Private message not received to the other client